

Algorithms Problem Solving

Level 3

Join String



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Definition

Delimited String Concatenation

-  Program combines multiple strings stored in a vector into one string.
-  A custom delimiter is inserted between consecutive elements.

Explanation

Vector Traversal

-  The function iterates through every element of the input vector.
-  Each token is added to the result string.

Delimiter Placement

-  The delimiter is inserted between elements rather than being part of the original tokens.
-  This reconstructs a single formatted string.

Result Construction

-  The final result contains all vector elements in their original order.
-  It represents the reverse direction of the tokenization performed by ``SplitString()``.

Procedures

- 1 Define `JoinString(vector<string>, delimiter)`.
- 2 Initialize an empty result string.
- 3 Traverse the vector.
- 4 Append each token and the required delimiter.
- 5 Remove or avoid the final delimiter.
- 6 Return the joined string.

💡 Example

Vector:

`{"Sara", "Alterkawi"}`

Delimiter:

`" "`
`,`

➡ Result

Sara, Alterkawi

Note

- ◆ `JoinString()` is conceptually the inverse of `SplitString()`.
- ◆ Delimiters can contain multiple characters.
- ◆ A delimiter should normally appear only between elements, not after the final element.

⊘ Warning

- ◆ Empty vectors must be handled before attempting to remove a trailing delimiter.
- ◆ Delimiter length must be considered when removing a final delimiter.
- ◆ The function should preserve the original order of vector elements.



Comparison

▨ `JoinString()`

- ✓ Converts multiple strings into one string.
- ✓ Inserts a chosen delimiter between tokens.
- ✓ Can reconstruct text after vector-based processing.

▤ `SplitString()`

- ✓ Converts one string into multiple tokens.
- ✓ Searches for delimiters to identify boundaries.
- ✓ Represents the opposite direction of the operation.

✦ Summary

- ✓ Traverses a vector of strings.
- ✓ Combines the elements into one result.
- ✓ Inserts a custom delimiter between tokens.
- ✓ Provides the inverse operation of string splitting.

Thank you 😊